

JEE MAIN REVISION MASTER: BINOMIAL THEOREM

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	General Term, Independent Term & Binomial Coefficients	9 PYQs	Binomial Expansion

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
General Term, Independent Term & Binomial Coefficients	$T_{r+1} = {}^nC_r x^{n-r} y^r$, $\sum_{r=0}^n {}^nC_r = 2^n$, $C_0 + C_2 + C_4 + \dots = 2^{n-1}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 General Term, Independent Term & Binomial Coefficients (High Yield — 9 PYQs)

Core Principle: Binomial expansion: $(a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$. General term $T_{r+1} = {}^nC_r a^{n-r} b^r$. Number of terms in expansion of $(a+b)^n$ is $n+1$. Middle term: If n is even, one middle term at $r = n/2$; if n is odd, two middle terms at $r = (n-1)/2$ and $r = (n+1)/2$. Sum of all coefficients = 2^n .

■■ **Caution / JEE Pitfall:** Remember that T_{r+1} gives the $(r+1)$ -th term, so for the 5th term you substitute $r = 4$, not $r = 5$!

■ **Speed Trick:** Sum of coefficients of any polynomial expansion $(ax+by)^n$ is obtained immediately by putting $x = 1, y = 1$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value at $x=2$ for $\frac{1}{x^3} + 3x^2 + 3x + 1x^4 + 4x^3 + 6x^2 + 4x + 1 \dots$ " [Ans: C]
- [2025 Shift] Q#2 (MCQ): "If the number of terms in the binomial expansion of $(2x+3)^{3n}$ is 22, then the value of n is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The value of ${}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3 + {}^{46}C_3 + {}^{45}C_3 + {}^{45}C_4$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value at $x=2$ for $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$

- (A) 3
- (B) $(25 / 61)$
- (C) $(1 / 3)$ **[Correct]**
- (D) $(19 / 73)$

Correct Answer: C

Solution Analysis: The given expression is $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$. We can recognize both numerator and denominator using identities. $x^3+3x^2+3x+1=(x+1)^3$ and $x^4+4x^3+6x^2+4x+1=(x+1)^4$. So the expression becomes $\frac{(x+1)^3}{(x+1)^4}$. Cancel the common factor $(x+1)^3$: $\frac{(x+1)^3}{(x+1)^4} = \frac{1}{x+1}$. Now put $x=2$: $\frac{1}{2+1} = \frac{1}{3}$. So the correct option is $(1 / 3)$ Option C.